NanoGap-Coupled Graded Gyroid Cube

Digital geometry, mathematical construction, and a proposed experimental benchmark

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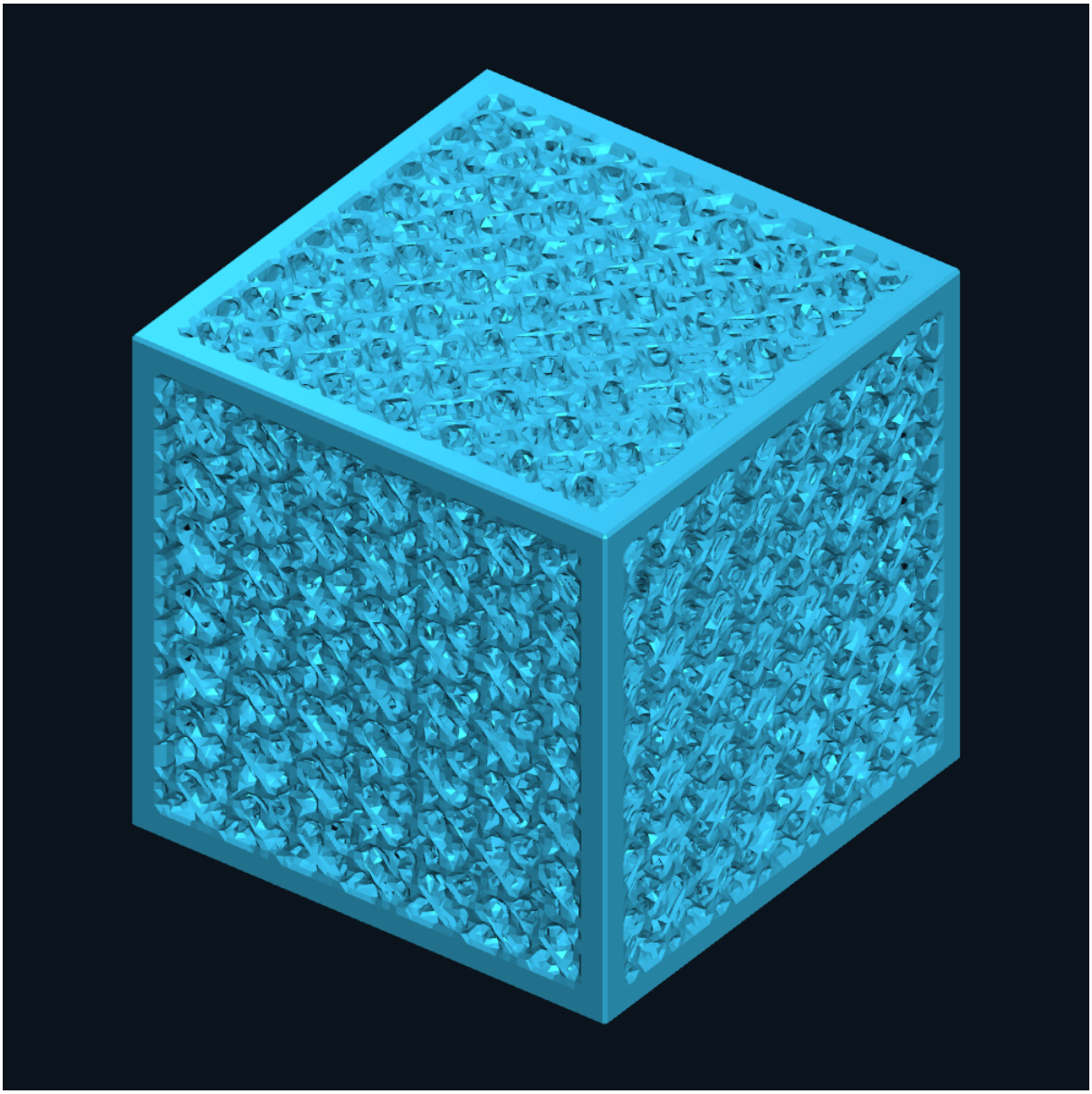


Figure 1. Rendering of the original high-resolution STL model. Digital geometry only; no printed prototype or measured strength is asserted.

Scope of submission. This research package proposes an independent feasibility and mechanical evaluation of an original July 2026 63.5-mm architected cube. It contains STL geometries, original mathematical source, design manifest and preflight audit. It is not a claim of structural superiority over the published MIT/Northeastern scale-rich network materials.

1. Research question and design provenance

Can a functionally graded gyroid sheet lattice with a tetrahedrally oriented internal support system and proximity-dependent geometric reinforcement provide useful stiffness, energy absorption, or damage tolerance at controlled material fraction, compared against matched reference lattices?

The original v1.0 model and source were documented in a July 2026 package and shared with an external recipient at that time. This document is an evaluation proposal, not a priority or patentability determination.

2. Digital specimen inventory

Model file	Purpose	Key observation
63.5-mm high-resolution master	Primary geometry	1,696,350 triangles; modeled volume 73.68 cm3
63.5-mm standard model	Inspection and slicing	672,786 triangles; modeled volume 59.22 cm3
38.1-mm calibration cube	Print feasibility	Not a mechanically equivalent full-scale sample
63.5-mm lattice core	Component visualization	Overlaps frame; not separate assembly
63.5-mm outer frame	Component visualization	Overlaps core; not separate assembly

All STLs use millimeters by convention; the STL format itself has no unit metadata. The separate lattice and frame are geometrically overlapping exports, not detachable physical parts.

3. Mathematical construction

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Cube-shell radius: r = max(|x|, |y|, |z|) / (L/2), with L = 63.5 mm.

Nominal pitch: P(r) = 8 + 6(1 - r^1.35) mm.

Threshold grading: tau(r) = 0.27 + 0.21 r^1.60.

Gyroid: G = sin(kx)cos(ky) + sin(ky)cos(kz) + sin(kz)cos(kx); k = 2 pi / P(r).

Proximity coupling: K(d) = exp(-0.22 d); tau_eff = tau(r) + 0.080 K(d).
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Solid field: |G| <= tau_eff, united with four tetrahedrally directed center struts, a central node, and a 12-edge frame. Distance d is measured in millimeters to the nearest strut; "NanoGap" is a project name, not a claim of nanoscale manufactured features.

4. Preflight results and currently unresolved issues

Preflight item	Result / limitation
Digital format	All five binary STL byte counts and facet totals internally consistent; models are geometry files, not test data
High-resolution envelope	Approximately 63.5 x 63.5 x 63.5 mm; 28.78% modeled solid fraction
Mesh convergence	Standard and master volume difference approx. 24.42% of standard model volume; must be reconciled before comparable experiments
Variable-phase interpretation	Implemented k(r)x phase does not have local wavelength equal to P(r) along radial-dominant directions
Physical validation	No load curves, independently printed specimen, calibrated material model, or measured impact performance supplied

A local x-direction phase analysis yields $P_{phase}(r) = P(r)^2 / [P(r) - r P'(r)]$ in a region where x controls the Chebyshev radius. At $r = 1$, P_{phase} is approximately 3.98 mm even though nominal $P(1) = 8$ mm. This concerns the phase field, not a directly measured pore diameter.

5. Interpretation relative to recent work

The MIT/Northeastern publication Scale-rich network-based metamaterials reports fabricated networks and controlled mechanical experiments. Its reported network tunability and impact performance are not direct comparative measurements against the present AXEZENT cube. It would be methodologically incorrect to rank the two architectures without matching density, material, geometry, boundary conditions and response variables.

6. Proposed collaboration

We respectfully request a feasibility review and, if practical, a protocol discussion for controlled mechanical testing. Suggested first priority: reproduce/repair geometric convergence and establish reliable specimens. Second priority: matched-density compression and impact comparisons with independent controls. Third priority: deliberately seeded damage, repeatability and optional calibrated numerical models.

7. Preliminary laboratory work plan

Phase A: Geometry and print readiness. Compare candidate mesh resolutions; assess connectedness, wall thickness, minimum gaps, slicer repairs, fabrication constraints, material conditioning and print process. Begin with the 38.1-mm calibration, then select one 63.5-mm converged full-size geometry. Record as-printed mass, dimensions, pore closure and defects.

Phase B: Controlled mechanics. Select a uniform reference lattice and, if feasible, a scale-rich design derived from the authors' released methodology. Match actual volume fraction and material, specimen envelope, printer process and loading orientation. Determine appropriate lab standards and specimen count (five independent samples per condition suggested only as an exploratory starting point).

Phase C: Metrics and robustness. Report stress-strain and force-displacement curves; define a failure criterion; measure stiffness, yield/crush response, energy absorbed per unit mass, peak impact transmission and variation. Apply reproducible notches/removed struts to evaluate failure localization and damage tolerance.

Phase D: Computation and dissemination. Where useful, fit calibrated finite-element material parameters and compare numerical predictions with experiments. Report negative or inconclusive results as well as improvements. Discuss ownership, attribution, access to raw results and publication arrangements before proceeding.

8. Data handoff and limitations

The accompanying archive includes all five original STL variants, the original Python generator and JSON parameter manifest, a numerical geometry receipt, an STL rendering, research references, and file checksums. It contains no finite-element solver output, no structural wall STL and no claim of tested nanogap-scale fabrication. A separate design phase is required for a wall-panel implementation.

The technical description and test proposal are supplied for academic evaluation. Any broader distribution, licensing, financial arrangement, coauthorship, or public release should be discussed explicitly between the researcher and recipient institutions. This notice by itself does not constitute a nondisclosure agreement.

9. Reference

Both, C., Chen, A. Y.-J., Gao, T.-T. et al. (2026). Scale-rich network-based metamaterials. Nature 659, 345-356. DOI: 10.1038/s41586-026-11191-2.

Original AXEZENT engineering package: AXZ_Dr_Shawn_Cube_v1_0_STL_Package.zip; original high-resolution master STL, dated July 2026; verified files are included in the accompanying research archive.